

INSTALLATION OF JAVA AND WEBGOAT

IT 323 - Information Assurance and Security (Lecture) Fri 11:30-1:00 PM

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EVIDENCE #1: JAVA ENVIRONMENT VERIFICATION

The initial phase of the activity involved verifying the Java environment. WebGoat requires Java 15 or greater to function correctly. By executing the `java -version` command in the terminal, we confirmed the presence of the OpenJDK Runtime Environment (Version 17). This ensures the machine is capable of running the Java Archive (.jar) server correctly.

```
junaiah_guarino@cloudshell:~$ java -version
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)
junaiah_guarino@cloudshell:~$
```

EVIDENCE #2: WEBGOAT SERVER ACTIVATION

PART I: Download and Run WebGoat

The WebGoat server was launched using the `java -jar` command. For this activity, we utilized Google Cloud Shell, which provides a secure, sandboxed environment.

```
HTTP request sent, awaiting response... 200 OK
Length: 118208649 (113M) [application/octet-stream]
Saving to: 'webgoat-2023.8.jar'

webgoat-2023.8.jar      100%[=====] 112.73M  13.4MB/s   in 7.0s

2026-04-15 11:57:04 (16.2 MB/s) - 'webgoat-2023.8.jar' saved [118208649/118208649]

junaiah_guarino@cloudshell:~$
```

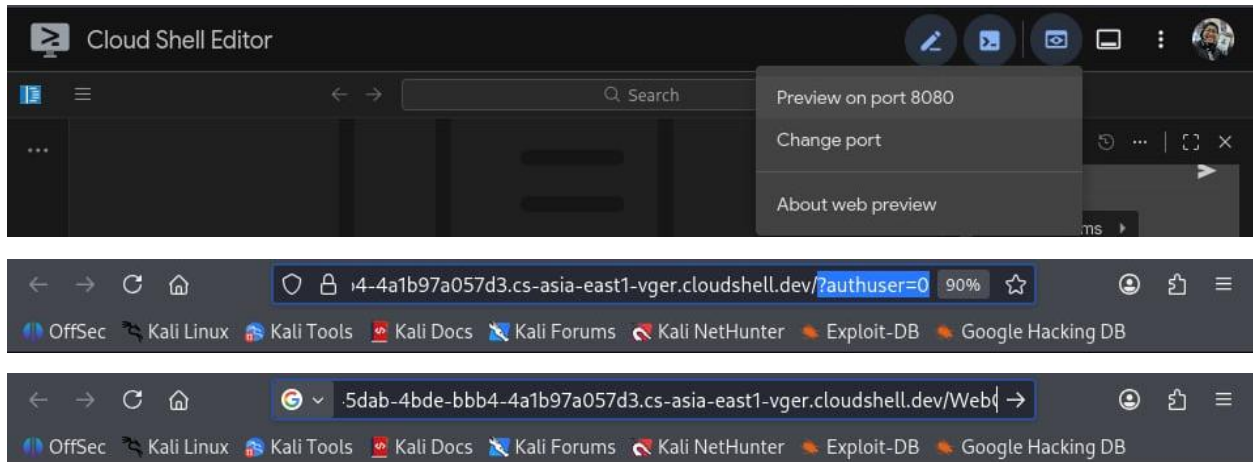
```
2026-04-15T11:58:00.787Z INFO 1581 --- [main] org.owasp.webgoat.server.StartWebGoat : Started StartWebG
oat in 2.448 seconds (process running for 3.854)

WebGoat

2026-04-15T11:58:00.906Z INFO 1581 --- [main] org.owasp.webgoat.server.StartWebGoat : No active profile
set, falling back to 1 default profile: "default"
2026-04-15T11:58:02.980Z INFO 1581 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spr
ing Data JPA repositories in DEFAULT mode.
2026-04-15T11:58:03.109Z INFO 1581 --- [main] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring D
ata repository scanning in 102 ms. Found 2 JPA repository interfaces.
2026-04-15T11:58:04.208Z WARN 1581 --- [main] io.undertow.websockets.jsr : UT026010: Buffer
pool was not set on WebSocketDeploymentInfo, the default pool will be used
2026-04-15T11:58:04.258Z INFO 1581 --- [main] io.undertow.servlet : Initializing Spring
embedded WebApplicationContext
2026-04-15T11:58:04.263Z INFO 1581 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicati
onContext: initialization completed in 3215 ms
2026-04-15T11:58:04.706Z INFO 1581 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - St
arting...
2026-04-15T11:58:05.367Z INFO 1581 --- [main] com.zaxxer.hikari.pool.PoolBase : HikariPool-1 - Dr
iver does not support get/set network timeout for connections. (feature not supported)
2026-04-15T11:58:05.371Z INFO 1581 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Ad
ded connection org.hsqldb.jdbc.JDBCConnection@443dbe42
2026-04-15T11:58:05.376Z INFO 1581 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - St
```

PART 2: Create a user account

1. Access was established by configuring the Web Preview on port 8080 and appending /WebGoat to the generated URL. This step creates a localized web application environment for security testing.



2. After successfully launching the WebGoat server, we opened the web browser to access the registration interface, where we created a local user account by providing a username and password to ensure our progress is stored within the application's database for the duration of the laboratory.

Register

Username

applebunny24

can only contain lowercase letters, digits, and -

Password

••••••••

Confirm password

••••••••

Terms of use

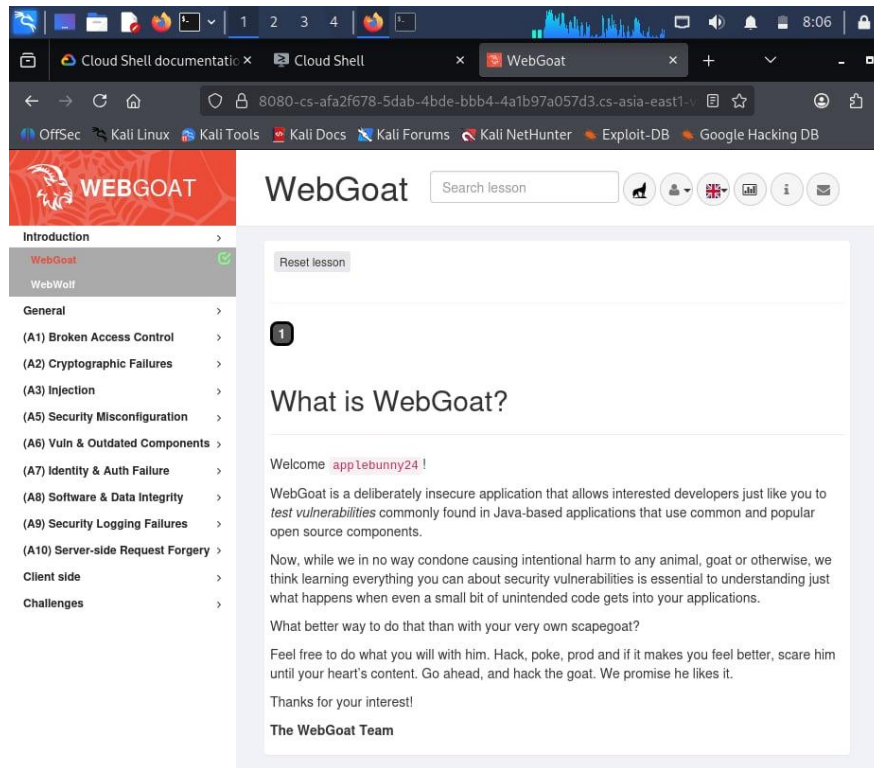
While running this program your machine will be extremely vulnerable to attack. You should disconnect from the Internet while using this program. WebGoat's default configuration binds to localhost to minimize the exposure.

This program is for educational purposes only. If you attempt these techniques without authorization, you are very likely to get caught. If you are caught engaging in unauthorized

☒ Agree with the terms and conditions

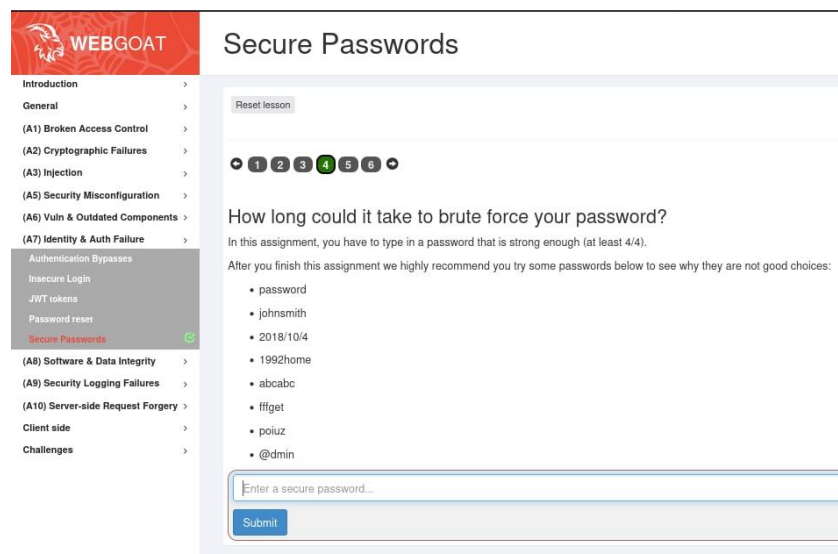
Sign up

3. The screenshot below shows the WebGoat home page after successfully logging in with the newly created account.



EVIDENCE #3: SECURE PASSWORD LESSON (STEP 4)

We navigated to the (A7) Identify and Auth Failures section (formerly Broken Authentication) to complete the Secure Passwords lesson. This assignment focuses on the concept of password entropy and the time required for a successful brute-force attack.



✓

09547899095ILY|

☐ Show password

Submit

You have succeeded! The password is secure enough.

Your Password: *****

Length: 14

Estimated guesses needed to crack your password: 30000100000000

Score: 4/4

Estimated cracking time: 95129 years 252 days 23 hours 6 minutes 40 seconds

Score: 4/4

We completed the introductory learning modules and successfully finalized the activity by entering a complex password that met all security requirements, resulting in the Step 4 indicator turning green to confirm completion.